UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #5

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: ________________________________
Due Date: Saturday April 04 by 12:00 noon

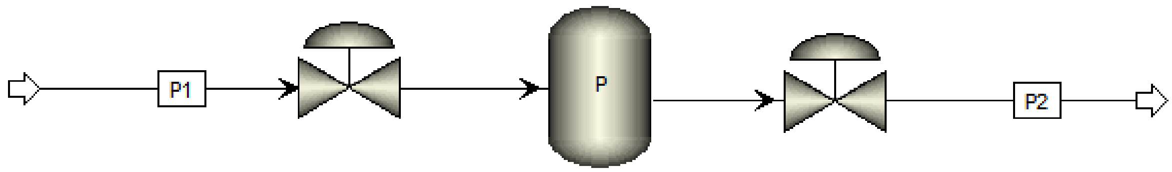
Number of points..........................100
Your score.................................

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment5_XY.PDF
  - X: First letter of your first name
  - Y: Your last name
  - Example: Assignment5_MCaracotsios.PDF
- Email your PDF file to: Daniel Christiansen dchris28@uic.edu and Yuechen Gao ygao54@uic.edu by the due date (Saturday April 04) and time (12:00 noon)

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 5A: Consider the gas storage tank in the process flow diagram below. The volume of the tank is V and the temperature T is constant.



The molar gas flow rates are determined by the following equations:

$$F_1 = x_1 C_{V1}\sqrt{P_1 - P}$$
$$F_2 = x_2 C_{V2}\sqrt{P - P_2}$$

where C_{V1} and C_{V2} are the valve coefficients and $\{x_1, x_2\}$ are the upstream and downstream valve openings. Assume the gas in the tank behaves as an ideal gas. Derive a transfer function model relating the tank pressure P to the inlet valve opening x_1. Develop expressions for the process gain K_P and lag time constant τ_P of the transfer function.

Problem 5B A gas stream containing 10 mol% ammonia is fed into an absorber to remove most of the ammonia before being vented to the atmosphere. This is being done by absorbing it with water. A step test is performed to investigate the response of the ammonia concentration in the treated gas to changes in the water flow rate. The table below shows the concentration of ammonia in ppm to a step decrease in the water flow rate.

Time, s	Water flow, gpm	Treated Gas NH_3, ppm
0	250	50.00
0	200	50.00
20	200	50.00
30	200	50.12
40	200	50.30
50	200	50.60
60	200	50.77
70	200	50.90
80	200	51.05
90	200	51.20
100	200	51.26
110	200	51.35
120	200	51.48
130	200	51.55
140	200	51.63
160	200	51.76
180	200	51.77
250	200	51.77

Reconstruct an approximate transfer function for the absorption process. Investigate the following models:

1. First order
2. First order with time delay
3. Second order
4. Second order with time delay

Explain the method that you use to reconstruct the transfer function for each model that you postulate.

Plot the data and the model responses and determine using the Akaike information criterion (AIC) which model best represents the absorption process.

Prepare a table that contains a summary of your results in the following format and highlight the best and worst models:

Model	K_p	τ_{p1}	τ_{p2}	θ	RSQ	AIC
1						
2						
3						
4						

RSQ: Residual Sum of Squares
AIC: Akaike Information Criterion